1. (a)
$$p(p|x) \propto p^{a-1}(1-p)^{b-1}(1-p)^{k-1}p = p^a(1-p)^{b-k-2},$$
which is the kernel of the Beta$(a+1, b-k-1)$-distribution. Hence, this is a conjugate pair.

   (b)
$$p(\mu|x) \propto \frac{\mathbb{1}_{[-1,1]}(\mu)}{2}\frac{1}{\sqrt{2\pi}}e^{-(x-\mu)^2/2}$$
is not a uniform distribution. Hence, this is not a conjugate pair.

   (c)
$$p(\beta|x) \propto \frac{b^a}{\Gamma(a)}\beta^{a-1}e^{-b\beta}\frac{\beta^x}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x} \propto \beta^{a+x-1}e^{-(b+x)\beta},$$
which is the kernel of the Gamma$(a+x, b+x)$-distribution. Hence, this is a conjugate pair.

2. Let $X = (X_1, \ldots, X_n)|\lambda \sim \text{Pois}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$. The posterior is proportional to
$$p(\lambda|X) \propto \lambda^{a-1}e^{-b\lambda}\prod_{i=1}^{n} e^{-\lambda}\lambda^{x_i} = e^{-(b+n)\lambda}\lambda^{a-1+\sum_{i=1}^n x_i}.$$
The right-hand side is the kernel of a Gamma$(a + \sum_{i=1}^n x_i, b+n)$-distribution, which verifies conjugacy. The mean of a Gamma$(a, b)$ distribution is $a/b$, so that the posterior mean is
$$\frac{a + \sum_{i=1}^n x_i}{b+n} = \frac{b}{b+n}\frac{a}{b} + \frac{n}{b+n}\bar{x}.$$
As the sample size $n$ increases, the posterior mean shifts from the prior mean to the sample mean.

3. Given i.i.d data $X = (X_1, \ldots, X_n) \sim \text{Exp}(\lambda)$, the log-likelihood and its second derivative are
$$\begin{aligned}\ell(\lambda|X) &= n\log(\lambda) - \lambda n\bar{X},\\ \ell''(\lambda|X) &= -n/\lambda^2.\end{aligned}$$
Since $\ell''(\lambda|X)$ is independent of $X$, the Jeffrey's prior density is
$$p(\lambda) \propto \sqrt{\mathbb{E}_\lambda\left[\frac{n}{\lambda^2}\right]} = \frac{\sqrt{n}}{\lambda}$$
on $\lambda > 0$. Its integral does not converge so the prior is improper. But the posterior is
$$\begin{aligned}p(\lambda|X) &\propto p(\lambda)L(\lambda|X) = \frac{\sqrt{n}}{\lambda}\lambda^n e^{-\lambda n\bar{X}}\\ &= \sqrt{n}\lambda^{n-1}e^{-\lambda n\bar{X}}.\end{aligned}$$
This is the kernel of a Gamma$(n, n\bar{X})$-distribution, and hence the posterior is proper. Alternatively, one can check directly that the integral of $p(\lambda|X)$ over $\lambda \in [0, \infty)$ converges.